

Performance Testing

Date	29 June 2026
Team ID	xxxxxx
Project Name	OptiCrop – Smart Agricultural Crop Recommendation System
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Custom Multi-threaded Python Performance Testing Harness (urllib, threading, psutil and time libraries)
Type of Testing	Load Testing, Stress Testing, Spike Testing
Target Module / API	/predict (POST recommendation engine) and / (web interface homepage)
Test Environment	Local Linux Workstation (Python 3.14, Flask Development Server, Random Forest ML Model)
Test Date	29 June 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Homepage Load Test	1	0.05 s	Homepage responds successfully with minimal latency.
2	Predict API Load Test	10	1.88 s	Prediction API handles concurrent requests with low response time.
3	Predict API Stress Test	50	6.18 s	Application remains stable under sustained concurrent load.
4	Predict API Spike Test	100	11.99 s	Application withstands sudden traffic spikes without request failures.

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Average Response Time	< 2 seconds	0.881 sec (880.89 ms)	Pass	Average response time remained well within the target across all scenarios.
2	Maximum Response Time	< 5 seconds	1.291 sec (1290.86 ms)	Pass	Peak latency during the 100-user spike test remained well below the acceptable limit.
3	Throughput	N.A	83.40 requests/sec	Pass	Achieved high throughput while maintaining zero request failures.
4	Error Rate	< 1%	0.00%	Pass	Successfully completed all 1,650 requests without any HTTP or connection failures.
5	CPU Utilization	< 80%	76.20%	Pass	CPU utilization remained below the predefined threshold during peak workload.
6	Memory Utilization	< 80%	1.60%	Pass	Memory usage remained consistently low, indicating efficient resource management.

Step 4: Observations & Analysis

Key Findings:

- The OptiCrop application demonstrated stable performance across all testing scenarios, successfully processing 1,650 requests without any failures.
- The average response time was 880.89 ms, while the maximum response time remained 1.29 seconds, both well within the predefined performance targets.
- The prediction engine maintained a 0.00% error rate even during the 100-user spike test, indicating reliable request handling.
- Peak throughput reached 83.40 requests per second, demonstrating efficient processing capability for concurrent prediction requests.
- Resource utilization remained efficient, with 76.20% CPU usage and only 1.60% memory utilization under peak load.

Bottlenecks Identified:

- As the number of concurrent users increased from 10 to 100, response latency increased due to the Flask development server's limited request-handling capability.
- The primary bottleneck was request queueing under heavy concurrency rather than machine learning inference, as the Random Forest model itself remained lightweight and fast.
- The application was tested using Flask's built-in development server, which is not optimized for high concurrency production workloads.

Optimization Steps Taken:

- The Random Forest model was serialized using a Pickle pipeline, allowing the trained model to be loaded once and reused for subsequent prediction requests.
- Efficient input validation and preprocessing were implemented to minimize unnecessary computation before model inference.
- The machine learning pipeline combines preprocessing and prediction into a single Scikit-learn Pipeline, reducing processing overhead and simplifying inference.
- Based on the performance analysis, the application is recommended to be deployed using a production-grade WSGI server such as Gunicorn with multiple worker processes (or asynchronous workers like Gevent) to improve scalability and reduce latency under heavy traffic.

Step 5: Screenshots / Evidence

```
A optiCrop main v3.14.5
>
19:13:24

A optiCrop main v3.14.5
> python src/model_training.py
19:14:03
Loading dataset from /home/rajan/optiCrop/data/Crop_recommendation.csv...

--- Training and Evaluating Models ---
Logistic Regression -> Accuracy: 0.9727 | Precision: 0.9740 | Recall: 0.9727 | F1-Score: 0.9725
K-Nearest Neighbors -> Accuracy: 0.9795 | Precision: 0.9804 | Recall: 0.9795 | F1-Score: 0.9793
Decision Tree -> Accuracy: 0.9795 | Precision: 0.9806 | Recall: 0.9795 | F1-Score: 0.9794
Random Forest -> Accuracy: 0.9955 | Precision: 0.9957 | Recall: 0.9955 | F1-Score: 0.9955

Best Model: Random Forest with Accuracy: 0.9955

Saved best model pipeline to /home/rajan/optiCrop/models/crop_recommendation_model.pkl

A optiCrop main v3.14.5 1s
>
19:14:16
```

Figure 1: model_training

```
> source venv/bin/activate 20:07:50

A optiCrop main ? v3.14.5
> python "5. Project Development Phase/src/run_perf_test.py" 20:07:57
Starting Flask server at /home/rajan/optiCrop/5. Project Development Phase/app.py with /home/rajan/optiCrop/venv/bin/python...
Server response code: 200
Connection successful! Running performance scenarios...

--- Running Scenarios ---

Scenario 1: Homepage Load Test
Virtual Users: 1 | Total Requests: 50
Duration: 0.05 seconds
Avg Response Time: 0.91 ms
Max Response Time: 1.42 ms
Throughput: 1080.40 req/sec
Error Rate: 0.00%

Scenario 2: Predict API Load Test
Virtual Users: 10 | Total Requests: 100
Duration: 1.88 seconds
Avg Response Time: 183.72 ms
Max Response Time: 878.42 ms
Throughput: 53.32 req/sec
Error Rate: 0.00%

Scenario 3: Predict API Stress Test
Virtual Users: 50 | Total Requests: 500
Duration: 6.18 seconds
Avg Response Time: 587.93 ms
Max Response Time: 696.07 ms
Throughput: 80.90 req/sec
Error Rate: 0.00%

Scenario 4: Predict API Spike Test
Virtual Users: 100 | Total Requests: 1000
Duration: 11.99 seconds
Avg Response Time: 1141.09 ms
Max Response Time: 1290.86 ms
Throughput: 83.40 req/sec
Error Rate: 0.00%

Stopping Flask server...

=== PERFORMANCE TEST SUMMARY ===
Overall Avg Latency: 880.89 ms
Overall Max Latency: 1290.86 ms
Overall Error Rate: 0.00%
Max CPU Observed: 76.20%
Max Memory Observed: 1.60%
Results saved to /home/rajan/optiCrop/5. Project Development Phase/perf_test_results.json
```

Figure 2: performance_test